

CSE321 Quiz 02**Time 20 minutes**

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Q1. You have a File System where there are a total 1100 blocks of data region [where data can be stored]. The Inode Structure you are implementing has 4 direct blocks, 2 single indirect blocks and 1 double indirect block. The Address space needed for each block is 4 bytes. The size of the block is 128 bytes. 4

- What will be the maximum size of the file in this system given the requirements.
- You are trying to store a file with the size of 1074. How much will the internal fragmentation cost

Q2. Suppose in your file system, the address space for each block and size of each block both are the same (8 Bytes). One INODE system has 71 direct blocks and another has 4 direct blocks and 17 triple indirect blocks. Which will hold the bigger file size? 3

Q3. Consider these two INODE Structure: 3

INODE 1: 3 direct, 2 single indirect, 1 double indirect block.

INODE 2: 2 direct, 2 double indirect block

Data block size = 128 bytes, data block address space 8 bytes. You are storing a file of size 4352 bytes in both versions. You want to access the last byte of the file. Which implementation will ensure that the least amount of disk access count (Disk Read) in this scenario? **Justify with mathematical demonstration.**